

— IE437 · DATA-DRIVEN DECISION MAKING AND CONTROL — LECTURE 4

Bayesian Optimization

Put the belief to work — act to learn, and to win

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The first policy in the course — a rule that turns a belief into an action

— HANDOFF

The handoff — belief that acts

Lectures 2 and 3 built a belief and reasoned inside it — passively. Now the belief has to choose.

Where we are — belief stops observing and starts choosing

STAGES	static ● — ○	MODEL	data-driven ○ — ●	AGENTS	single agent ● — ○
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MODEL-BASED		DATA-DRIVEN
Static, single	optimisation (<i>Lec 1</i>)	Bayesian statistics · network (<i>Lec 2–3</i>) → Bayesian optimisation (<i>Lec 4</i>)

Nothing on the cube moves this lecture. We stay in the static, data-driven, single-agent cell and [go deeper into it](#): the belief we built in Lectures 2 and 3 stops being a report on the world and starts being the thing that decides what to do next.

The loop closes. Use the belief to choose a query, observe the answer, update the belief, repeat. That is a different kind of object from anything in Part I — not an answer but a [rule for producing answers](#), which is what the rest of the course will call a *policy*.

The setting — unknown, expensive, and every query counts

Lecture 1 minimised an f that was written down, cheap, and differentiable. Delete all three assumptions.

$$x^* = \arg \max_x f(x), \quad f \text{ unknown and expensive}$$

WHAT "UNKNOWN" COSTS YOU

No formula, so no gradient. No gradient, so none of Lecture 1's machinery applies directly — the only thing you may do with f is **ask it a question and receive a number**.

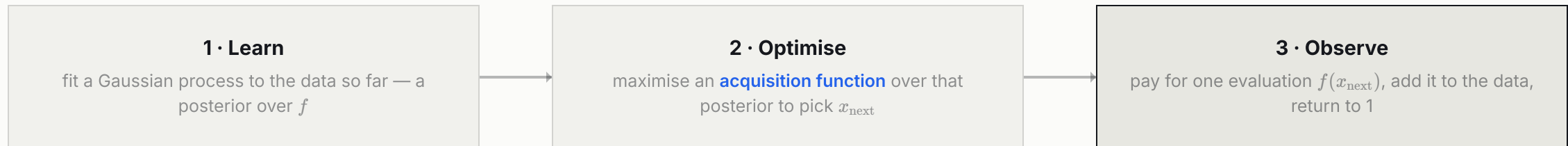
WHAT "EXPENSIVE" COSTS YOU

One evaluation is a wet-lab experiment, a multi-hour CFD run, a clinical trial, a season of a wind farm. You get **tens** of queries, not millions.
Every query must be spent deliberately.

Both halves matter. Unknown alone would be handled by fitting a model once and optimising it — that is Lecture 5. Expensive alone would be handled by a careful experimental design fixed in advance. It is *unknown and expensive together* that forces the loop: you must learn f and optimise it **at the same time**, with the same queries.

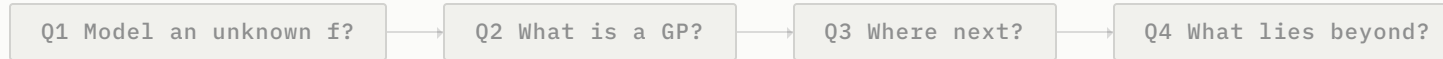
The thesis — model the unknown, then act on the model

Build a belief over the unknown function, then choose where to look by [balancing learning against winning](#).



Three earlier lectures fused into one turn of a crank: step 1 is Lecture 2's Bayesian update, now over a whole function; step 2 is Lecture 1's optimisation, now over a *cheap* surrogate; step 3 is the single expensive query we are trying to spend wisely.

The roadmap — four questions



- **Q1 — How do we model a function we have never seen?** A probabilistic **surrogate**: not one fitted curve but a distribution over the curves consistent with the data.
- **Q2 — What is that model, concretely?** The **Gaussian process** — and its one real assumption, the kernel.
- **Q3 — Where do we look next?** The **acquisition function**, which fuses "probably good" with "worth learning" into one optimisable score.
- **Q4 — What lies beyond?** Contextual BO, and the **bandit-to-RL** bridge.

— ACT 1

A belief over an unknown function

Q1. You cannot afford to probe f everywhere. So carry a distribution over the functions it might be.

The surrogate — a distribution, not a fitted curve

Q1 Model an unknown f ?

Q2 What is a GP?

Q3 Where next?

Q4 What lies beyond?

From a handful of evaluations, build a **surrogate**: a cheap stand-in for f that can be evaluated anywhere. The usual move is to fit one curve. The Bayesian move is to keep **all the curves the data has not ruled out**.

A FITTED CURVE GIVES YOU

one number at every x — a prediction. Where the data is dense and where it is absent, it looks equally confident. It cannot tell you where it is guessing.

A DISTRIBUTION GIVES YOU

a number **and a width** at every x . Narrow where data is dense, wide where data is sparse, and returning to the prior width far from everything.

That width is not decoration. It is **the signal we will use to decide where to look next**.

A function on a finite domain is a vector

The idea that makes a distribution over functions ordinary

Let $\mathcal{X} = \{x_1, \dots, x_n\}$ be finite, and let $\mathcal{H}$ be every function from $\mathcal{X}$ to $\mathbb{R}$. One such function is just a list of values —

$$f_0(x_1) = 5, \quad f_0(x_2) = 2.3, \quad f_0(x_3) = -7, \quad \dots, \quad f_0(x_n) = 8$$

— so f_0 is the vector $\mathbf{f} = [f(x_1), \dots, f(x_n)]^\top \in \mathbb{R}^n$, and a probability distribution over $\mathcal{H}$ is nothing more exotic than **a probability distribution over that vector**.

Take the simplest one, $\mathbf{f} \sim \mathcal{N}(\mu, \sigma^2 I)$, and the density over functions is written out in full:

$$p(\mathbf{f}) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2} (f(x_i) - \mu_i)^2\right)$$

The only thing wrong with this is the *independence*: it says knowing $f(x_1)$ tells you nothing about $f(x_2)$, however close they are. Replace $\sigma^2 I$ with a covariance matrix that couples nearby inputs, let $n \rightarrow \infty$, and you have a Gaussian process. Everything after this slide is that one substitution.

Lecture 2's posterior, with the parameter replaced by a function

The handoff from Chapter 2 is not an analogy. It is the *same five steps*, with one symbol swapped.

STEP	LECTURE 2 — BELIEF OVER θ	LECTURE 4 — BELIEF OVER f
Model	a likelihood family for y	$y_i = f(x_i) + \epsilon_i$
Prior	$p(\theta)$ — Beta, Gamma, Normal	$p(f) = \mathcal{GP}(m(\cdot), k(\cdot, \cdot))$
Likelihood	$p(y \mid \theta)$	$p(\mathbf{y} \mid \mathbf{f}) = \mathcal{N}(\mathbf{f}, \sigma_\epsilon^2 I)$
Posterior	$p(\theta \mid y)$ — conjugate, closed form	$p(f \mid \mathcal{D})$ — Gaussian, closed form
Predict	$p(\hat{y} \mid y) = \int p(\hat{y} \mid \theta) p(\theta \mid y) d\theta$	$p(f^* \mid x^*, \mathcal{D}) = \int p(f^* \mid x^*, \mathbf{f}) p(\mathbf{f} \mid \mathcal{D}) d\mathbf{f}$

Chapter 2 ended on the Normal–Normal case and noted that the predictive variance splits into measurement noise you can never remove plus parameter uncertainty you can. **Replace the parameter by an entire function and that same predictive integral is Gaussian-process regression.** The conjugacy that made Chapter 2 computable is the *only* reason Chapter 4 is computable too.

— ACT 2

The Gaussian process

Q2. A prior over functions whose posterior is available in closed form — and one free choice, the kernel, which carries every assumption you are making.

The Gaussian process — a prior over functions

Q1 Model an unknown f ?

Q2 What is a GP?

Q3 Where next?

Q4 What lies beyond?

A **Gaussian process** is a collection of random variables $\{f(x) : x \in \mathcal{X}\}$ such that **every finite subcollection is jointly Gaussian**. Written out, for any $x_1, \dots, x_n$:

$$\begin{bmatrix} f(x_1) \\ \vdots \\ f(x_n) \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} m(x_1) \\ \vdots \\ m(x_n) \end{bmatrix}, \begin{bmatrix} k(x_1, x_1) & \cdots & k(x_1, x_n) \\ \vdots & \ddots & \vdots \\ k(x_n, x_1) & \cdots & k(x_n, x_n) \end{bmatrix} \right), \quad f(\cdot) \sim \mathcal{GP}(m(\cdot), k(\cdot, \cdot))$$

THE MEAN FUNCTION

$m(x) = \mathbb{E}[f(x)]$ — the overall trend. Taken as $m \equiv 0$ almost always: it costs no expressiveness and saves computation.

THE COVARIANCE FUNCTION

$k(x, x') = \mathbb{E}[(f(x) - m(x))(f(x') - m(x')))]$ — how strongly nearby points are tied together. It must make every $\mathbf{K}$ positive semidefinite. **This is where all the modelling lives.**

A function drawn from a GP prior is, in the source's own phrase, *an extremely high-dimensional vector drawn from an extremely high-dimensional multivariate Gaussian*. It is Lecture 3's graph over a **continuum** of variables, with the kernel standing in for the edge structure.

Conditioning is the whole of it

Draw $\mathbf{f} = [f_1, \dots, f_{25}] \sim \mathcal{N}(0, \mathbf{K})$ and plot the 25 numbers in order: they look like a smooth curve, because $\mathbf{K}$ made neighbouring entries nearly identical. Read two entries of $\mathbf{K}$ off the diagonal band:

NEIGHBOURS — $\text{CORR}(F_1, F_2) = 0.966$

The joint $p(f_1, f_2)$ is a thin cigar. Observe $f_1 = -0.313$ and the conditional $p(f_2 | f_1)$ is **a narrow spike** — you have almost measured f_2 without paying for it.

FOUR APART — $\text{CORR}(F_1, F_5) = 0.573$

The joint is a fat ellipse. The same observation $f_1 = -0.313$ leaves $p(f_5 | f_1)$ **barely narrower than the prior** — that value you still have to buy.

And the mechanism is a single fact of linear algebra, the one the probability appendix calls Property 4:

$$\begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}\right) \implies Y_2 | Y_1 = y \sim \mathcal{N}(\mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(y - \mu_1), \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12})$$

The kernel decides how much one observation is worth at every other point in the domain. Everything the GP does — the narrowing, the interpolation, the uncertainty that grows away from data — is that one formula applied at scale.

GP regression — mean and uncertainty, in closed form

Observe $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ with $y_i = f_i + \epsilon_i$, $\epsilon_i \sim \mathcal{N}(0, \sigma_\epsilon^2)$. Prior and likelihood are Gaussian, so the joint of the data and the value at any new x is Gaussian:

$$\begin{bmatrix} \mathbf{y}_{1:n} \\ f \end{bmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{bmatrix} \mathbf{K} + \sigma_\epsilon^2 \mathbf{I} & \mathbf{k} \\ \mathbf{k}^\top & k(x, x) \end{bmatrix}\right)$$

Condition on $\mathbf{y}_{1:n}$ and read off $p(f \mid \mathcal{D}) = \mathcal{N}(\mu(x \mid \mathcal{D}), \sigma^2(x \mid \mathcal{D}))$ with

$$\mu(x \mid \mathcal{D}) = \mathbf{k}^\top (\mathbf{K} + \sigma_\epsilon^2 \mathbf{I})^{-1} \mathbf{y}_{1:n}, \quad \sigma^2(x \mid \mathcal{D}) = k(x, x) - \mathbf{k}^\top (\mathbf{K} + \sigma_\epsilon^2 \mathbf{I})^{-1} \mathbf{k}$$

The GP hands us, at every x , both a best guess *and* **how much to trust it** — and that pair is exactly what the next decision needs.

Read the two formulas. The mean is a **kernel-weighted average of the observed y 's** — a linear smoother whose weights the kernel chose. The variance starts at the prior value $k(x, x)$ and is reduced by $\mathbf{k}^\top (\cdot)^{-1} \mathbf{k}$, a quantity that is large near data and vanishes far from it. Note that σ^2 **does not depend on y at all**: where the GP is uncertain is fixed the moment you choose *where* to look, before you see a single answer. That is what makes experimental design possible.

The kernel is the assumption — and the data can pick it

A SMALL VOCABULARY

- **Squared exponential** $k = \sigma_0^2 \exp\left[-\frac{1}{2}\left(\frac{x-x'}{\lambda}\right)^2\right]$ — stationary, infinitely differentiable, *very* smooth. λ is the length scale, σ_0 the amplitude.
- **Matérn** $\frac{3}{2}, \frac{5}{2}$ — finitely differentiable, visibly rougher sample paths, usually more honest about physical data.
- **ARD** — one λ_d per input dimension; a large λ_d means dimension d is **irrelevant**.
- **Sums and products** — $k_1 + k_2$ adds independent processes, $k_1 k_2$ multiplies them: trend plus periodicity, periodicity with growing amplitude.

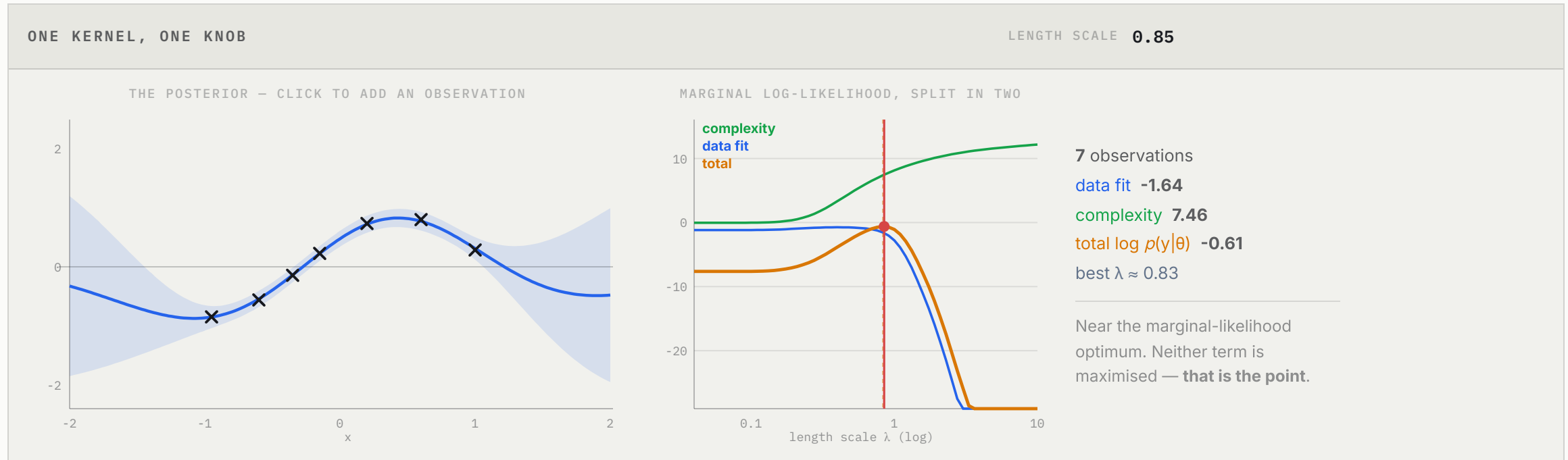
FITTING $\theta = (\Sigma_E, \Sigma_0, \mathbf{A})$

Maximise the **marginal likelihood** of the data:

$$\theta^* = \arg \max_{\theta} \left[\underbrace{-\frac{1}{2} \mathbf{y}^\top (\mathbf{K}_\theta + \sigma_\epsilon^2 \mathbf{I})^{-1} \mathbf{y}}_{\text{data fit}} \underbrace{-\frac{1}{2} \log |\mathbf{K}_\theta + \sigma_\epsilon^2 \mathbf{I}|}_{\text{complexity}} \right] - \frac{n}{2} \log 2\pi$$

The first term rewards explaining the data, the second rewards a *rigid* model. Their sum peaks in between — **an Occam balance, automatic and free**, and itself an optimisation (Lecture 1) sitting inside the loop.

Turn the dial and watch the assumption move



The same seven observations, one kernel, one knob. Short length scale: the posterior spikes at each datum and falls back to the prior between them — the model believes nothing carries. Long length scale: a near-straight line that cannot bend to the data. The right panel is the marginal likelihood split into its two terms, and **the total peaks where neither term is happy** — that is the Occam balance, drawn.

— ACT 3

Where to look next

Q3. The posterior gives a best guess and a width at every x . One number has to fuse them, and that number is a rule for acting.

The acquisition function — a score you are allowed to optimise

Q1 Model an unknown f ?

Q2 What is a GP?

Q3 Where next?

Q4 What lies beyond?

Given the posterior (μ, σ) , the **acquisition function** $A(x)$ scores how worthwhile it would be to query x next. Then

$$x_{\text{next}} = \arg \max_x A(x)$$

which is a *cheap* inner optimisation over the surrogate — no expensive evaluations, gradients available, run it as long as you like.

TWO IMPULSES

High $\mu(x)$ says *probably good* — spend the query where the model already expects a win. That is **exploitation**.

High $\sigma(x)$ says *worth learning* — spend the query where the model is ignorant, because that is where a surprise can hide. That is **exploration**.

ONE NUMBER

$A(x)$ is whatever function of μ and σ you are willing to defend. The three standard answers differ only in **how they weigh the two impulses**, and the disagreement is real: on the same posterior they point at different places.

This is the explore–exploit dilemma — the same one inside every reinforcement learner — but here it is not a heuristic. It is an explicit, differentiable, optimisable score, which is why Bayesian optimisation is the cleanest place in the course to meet it.

Three scores

	RULE	READS AS	LEANS
Probability of improvement	$\text{PI}(x) = \Phi\left(\frac{\mu(x) - f^+ - \xi}{\sigma(x)}\right)$	<i>how likely</i> is any improvement at all	exploit
Expected improvement	$\text{EI}(x) = \mathbb{E}[\max(0, f(x) - f^+)]$	how likely and by how much	balanced
Upper confidence bound	$\text{UCB}(x) = \mu(x) + \kappa \sigma(x)$	optimism, at an explicit price κ	tunable

where $f^+ = \max_{x_i \in x_{1:t}} f(x_i)$ is the best value seen so far.

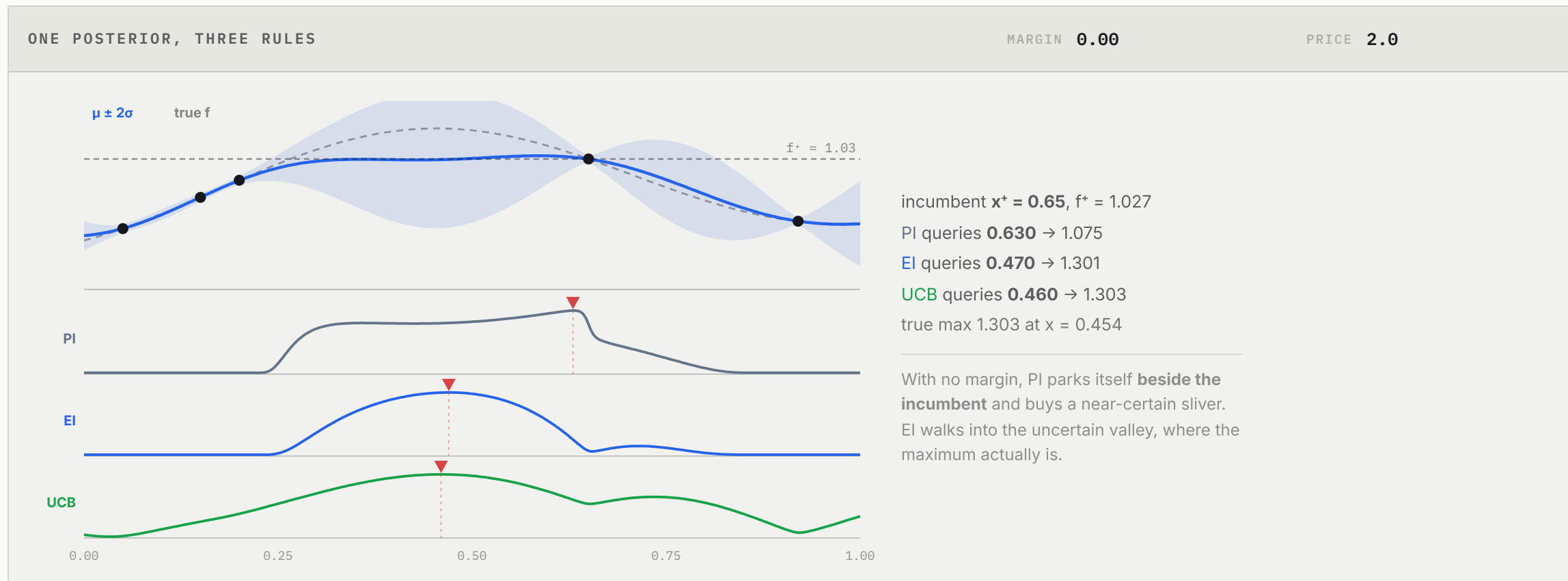
PI'S FLAW, AND WHY EI EXISTS

the source deck's own words

"The formulation is pure exploitation. Points that have a high probability of being **infinitesimally** greater than $f(x^+)$ will be drawn over points that offer larger gains but less certainty."

The patch is a margin $\xi \geq 0$ — demand improvement *by at least* ξ — scheduled large early and decayed to zero. But it is a knob you must tune: too small and the search is highly local, too large and it is excessively global. EI needs no such knob, because integrating $\max(0, f - f^+)$ already counts **how far above the line** the improvement is, not merely whether it happens.

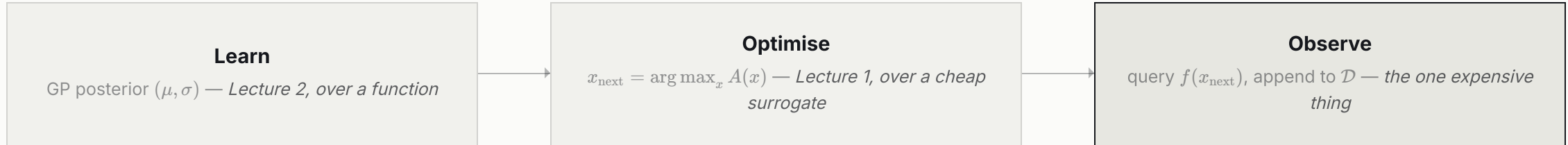
The three rules, disagreeing



One posterior, five observations, three scores drawn underneath it, each with its own $\arg \max$ marked. At $\xi = 0$, PI points at $x = 0.630$ — hard against the incumbent at 0.65, buying a near-certain sliver. EI points at $x = 0.470$, into the wide-uncertainty valley where the true maximum actually is. Turn ξ up and PI walks out to meet EI; turn κ down and UCB collapses onto the greedy mean.

The knob is the same knob in all three.

The loop

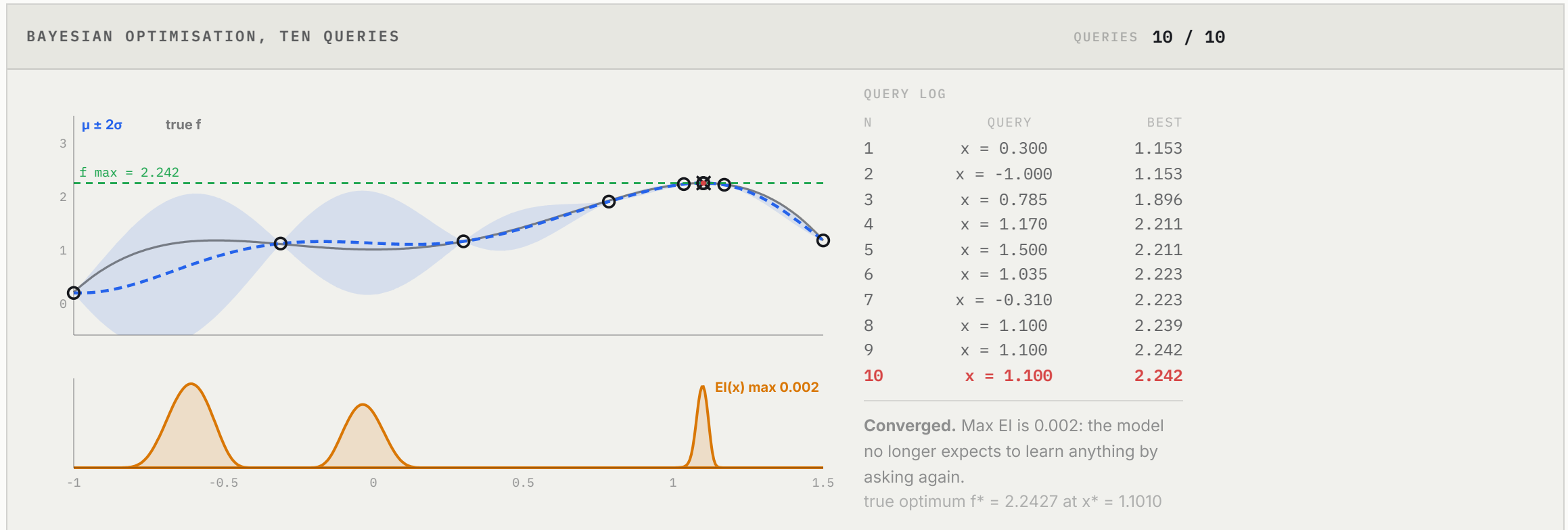


The dashed return arrow is what makes this a lecture rather than a technique. Each pass changes the belief, the changed belief changes the acquisition surface, and the changed surface changes where you look. **Nothing in Part I had a return arrow.**

Note also what the loop does *not* do: it never touches f except at step 3. All the optimisation happens on the surrogate. The whole design is an accounting trick for spending a scarce resource — expensive evaluations — by substituting a cheap one.

Ten queries on a quartic

Example 4.1 · maximise $-1.3x^4 + x^3 + 1.5x^2 + 1$ over $-1 \leq x \leq 1.5$ with noise $\sigma_\epsilon = 0.01$



Press *next query* and watch EI decide. The second query goes straight to the far boundary $x = -1$ — the mean there is unremarkable, but the uncertainty is enormous, and EI pays to find out. By the eighth the queries have collapsed onto $x = 1.10$, and the EI peak has fallen from 0.48 to 0.002: **the model no longer expects to learn anything by asking again**. The true maximum is $x^* = 1.1010$, $f^* = 2.2427$.

The lecture's own run took eleven queries and stopped at $x = 1.11$, $y = 2.24$ — the same answer, to the precision the noise allows.

This loop is a policy — the first in the course

THE BANDIT, LECTURE 4

$$\pi : [(a_1, r_1), \dots, (a_{t-1}, r_{t-1})] \rightarrow a_t$$

Find π^* maximising $\mathbb{E}[\sum_t r_t]$.

BAYESIAN OPTIMISATION, LECTURE 4

$$\pi : [(x^1, y^1), \dots, (x^{n-1}, y^{n-1})] \rightarrow x^n$$

Find π^* maximising $\mathbb{E}[\sum_t y^t]$.

The same definition twice: a rule that maps **the whole history to the next action**. That is what a policy is, and this is where the course meets one.

So say plainly what the acquisition function is. It is not a scoring heuristic bolted onto a regression — it is a **policy over a belief state**, and every policy in Part IV is its descendant. Lecture 8's ε -greedy rule chooses $\arg \max_a Q(a)$ with probability $1 - \varepsilon$ and explores otherwise; UCB there is $\mu_i + \sqrt{2 \ln t / n_i}$, which is $\mu + \kappa \sigma$ with the count standing in for the width. **Change the belief from a GP to a Q-table and the acquisition function is the exploration rule.**

— ACT 4

Beyond, toward RL

Q4. Strip the function away and the loop is a bandit. Add a context and it is a contextual bandit. Add dynamics and it is reinforcement learning.

Bayesian optimisation is a bandit with infinitely many arms



Put a slot machine under every point of the domain. Pulling arm x pays $f(x)$ plus noise; the payouts are unknown; you have a budget of pulls. That is the n -armed bandit with $n = \infty$ — and the reason it is not hopeless is the [kernel](#), which makes pulling one arm tell you about its neighbours.

WHAT THE BANDIT CONTRIBUTES

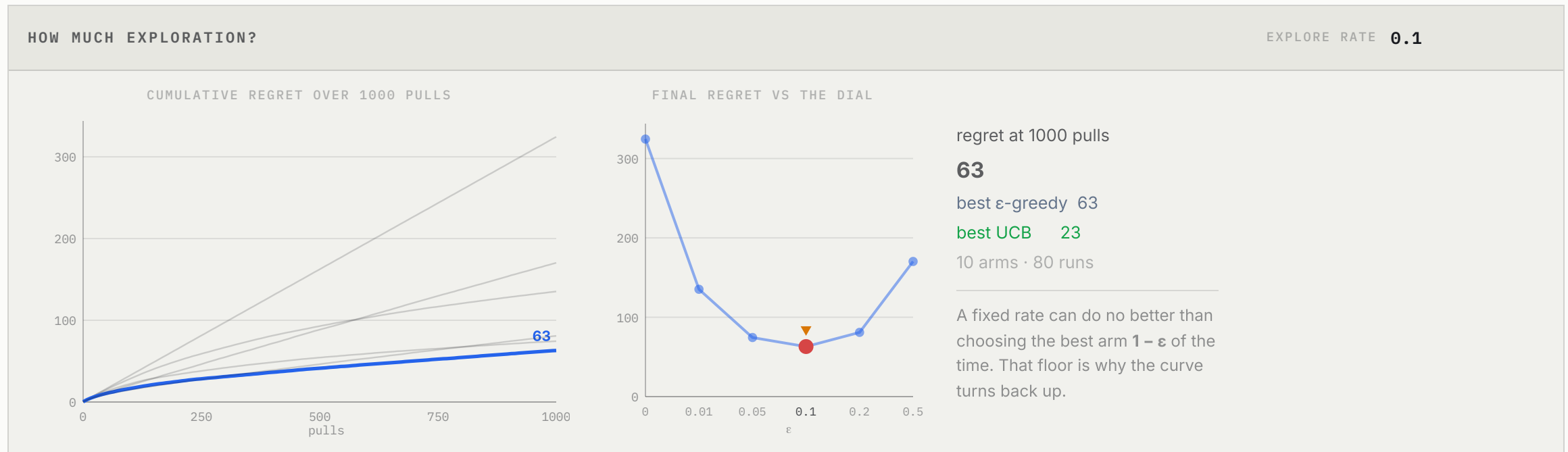
The word *policy*, and the trade-off in its bare form: **acquiring new information** against **capitalising on the information already held**. With finitely many arms the belief is one number per arm; here it is a whole GP.

WHAT BO CONTRIBUTES

Structure. A finite bandit must try every arm at least once, so its regret scales with the number of arms. A GP over a continuum needs only enough queries to pin down f at [the kernel's resolution](#), which is why tens of evaluations can suffice.

The lineage runs both ways. The bandit's Bayesian form is Chapter 2 exactly: after w wins and l losses, arm i has posterior $\text{Beta}(1 + w_i, 1 + l_i)$ and mean $\rho_i = \frac{w_i+1}{w_i+l_i+2}$ — the pseudo-count update from Lecture 2, now serving as [a belief state that an action will change](#).

How much exploration is the right amount?



Ten arms, unknown payout probabilities, a thousand pulls, cumulative regret on the vertical axis. Pure greed ($\epsilon = 0$) locks onto whichever arm happened to pay first and never recovers. Constant thrashing ($\epsilon = 0.5$) pays a fixed toll on every round. **The best fixed rate is in between, and the confidence-bound rule beats every fixed rate** — because it explores where the uncertainty actually is, rather than at random. That is the whole argument for $\mu + \kappa\sigma$, made without a Gaussian process anywhere in sight.

From a function to a context

Often the right action depends on a **context** c revealed just before each decision — the best price given the season, the best treatment given the patient, the best yaw angles given the wind direction. The object we want is no longer a point but a map:

$$x^* = \pi^*(c) = \arg \max_x f(x; c)$$

HOW THE GP ABSORBS A CONTEXT

Put a kernel on the context too and multiply:

$$k((x, c), (x', c')) = k_X(x, x') \cdot k_C(c, c')$$

which asserts that **similar contexts have similar optima**. Without that, each context would be a separate problem and no policy could be learned at all.

WHAT WE HAVE JUST BUILT

A rule that at each round takes all past (c, x, y) , maintains a belief state $B_t(f)$ over the unknown function, and acts to maximise cumulative reward.

That is an **MDP over the belief state** — Lecture 3's decision network, unrolled in time.

And it costs something up front. In the professor's own wind-farm study, the contextual learner starts at 0.79 average power efficiency against a greedy controller's 0.93, crosses it at roughly 2 500 iterations, and finishes ahead at 0.94. **Exploration is a debt you take on early and are repaid for later**, which is exactly the shape of every learning curve in Part IV.

The bridge — one table, four lectures

MODEL KNOWN — ONLY EXPLOITATION		MODEL UNKNOWN — EXPLORE VS EXPLOIT
Single state (<i>optimum action</i>)	optimisation (<i>Lec 1</i>)	bandit · Bayesian optimisation (<i>Lec 4</i>)
Multiple states (<i>optimum policy</i>)	dynamic programming (<i>Lec 7</i>)	contextual bandit → reinforcement learning (<i>Lec 8, 10</i>)

Read across the bottom row and you have Part IV. Read down the right column and you have this course's central axis. The one missing ingredient between a contextual bandit and full RL is **state dynamics**: in BO your action does not change the world it is asking about. [Let the action move the state, and the bandit becomes reinforcement learning.](#)

What carries over intact is the explore–exploit machinery of this lecture. What is added is a **value function**, to account for the future an action unlocks and not merely the reward it returns. That is Lecture 7.

— CLOSING

Closing

Part II is complete: belief built, belief structured, belief put to work.

Where we are — Part II complete

MODEL-BASED		DATA-DRIVEN
Static, single	optimisation (Lec 1 ✓)	Bayesian stats · network (Lec 2–3 ✓) → Bayesian optimisation (Lec 4 ✓)

What this lecture hands on is [the acquisition policy — the seed of an RL policy — and the bandit](#). Two roads lead out from it.

- **Lectures 5–6** — what if you [cannot query \$f\$ at all](#), and hold only a fixed dataset? Offline design optimisation, forward and inverse. Bayesian optimisation minus the loop.
- **Lectures 7–10** — what if the action [moves the world](#)? The loop stays, a value function is added, and the bandit becomes reinforcement learning.

Honest limits, before you go: a GP costs $O(n^3)$, so BO lives in the low thousands of evaluations, and the acquisition surface becomes flat and hostile in high dimension — in practice BO optimises **10–20 parameters**, not 10 000. Latent-space and neural surrogates push on both walls, and the first of them, learning a low-dimensional representation to optimise inside, is Lecture 6.

A known objective can be optimised; an unknown,
expensive one must be learned while it is
optimised.

A Gaussian process for the belief, an acquisition function to weigh learning against winning, and a loop that
— quietly, three lectures before the words appear — is the first reinforcement learner in the course.

Questions?

The acquisition function is a policy over a belief state. Everything Part IV does is add dynamics to that sentence.

— APPENDIX

Backup slides

Derivations and the bandit toolkit, kept out of the narrative.

Backup 1 — the GP posterior, from one Gaussian fact

The fact. If $\begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}\right)$, then

$$Y_2 \mid Y_1 = y \sim \mathcal{N}(\mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(y - \mu_1), \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12})$$

The application. Take $Y_1 = \mathbf{y}_{1:n}$ (observed, with σ_ϵ^2 on the diagonal) and $Y_2 = f(x)$ (the value we want). Then $\Sigma_{11} = \mathbf{K} + \sigma_\epsilon^2 \mathbf{I}$, $\Sigma_{21} = \mathbf{k}^\top$, $\Sigma_{22} = k(x, x)$, and with $m \equiv 0$:

$$\mu(x \mid \mathcal{D}) = \mathbf{k}^\top (\mathbf{K} + \sigma_\epsilon^2 \mathbf{I})^{-1} \mathbf{y}, \quad \sigma^2(x \mid \mathcal{D}) = k(x, x) - \mathbf{k}^\top (\mathbf{K} + \sigma_\epsilon^2 \mathbf{I})^{-1} \mathbf{k} \quad \blacksquare$$

Three readings. The mean is a **kernel-weighted average** of the observed y 's. The variance shrinks near data and returns to the prior $k(x, x)$ far from it. And the variance never sees $\mathbf{y}$ — only where you looked, not what you found — which is why a GP can plan an experiment before running it. The cost is the inverse: $O(n^3)$ once, $O(n^2)$ per prediction, which is the entire reason BO stops at a few thousand evaluations.

Backup 2 — kernels, and hyperparameters by marginal likelihood

Squared exponential. $k(x, x') = \sigma_0^2 \exp(-\frac{1}{2}\|x - x'\|^2/\lambda^2)$ — stationary, infinitely differentiable. **Matérn $\frac{3}{2}$:** $\alpha(1 + \sqrt{3}r)e^{-\sqrt{3}r}$; **Matérn $\frac{5}{2}$:** $\alpha(1 + \sqrt{5}r + \frac{5}{3}r^2)e^{-\sqrt{5}r}$, with $r = \|x - x'\|_2/l$ — finitely differentiable, rougher, usually more realistic. **ARD:** one λ_d per dimension, $k = \sigma_0^2 \exp[-\frac{1}{2} \sum_d ((x_d - x'_d)/\lambda_d)^2]$, and a large λ_d declares dimension d irrelevant. **Algebra:** $k_1 + k_2$ is the sum of independent processes, $k_1 k_2$ their product — so Lin + Per is *periodic with a trend*, Lin $\times$ Per is *growing amplitude*.

Fitting $\theta = (\sigma_\epsilon, \sigma_0, \lambda)$. Marginalise the latent $\mathbf{f}$ away and maximise what is left:

$$\theta^* = \arg \max_{\theta} \log \int p(\mathbf{y} \mid \mathbf{f}, \theta) p(\mathbf{f} \mid \theta) d\mathbf{f} = \arg \min_{\theta} \left[\frac{1}{2} \mathbf{y}^\top (\mathbf{K}_\theta + \sigma_\epsilon^2 \mathbf{I})^{-1} \mathbf{y} + \frac{1}{2} \log |\mathbf{K}_\theta + \sigma_\epsilon^2 \mathbf{I}| \right]$$

The two terms pull opposite ways as the length scale grows. On the seven-point example of the Act 2 widget, the complexity term $-\frac{1}{2} \log |\mathbf{K} + \sigma_\epsilon^2 \mathbf{I}|$ climbs monotonically from -0.04 at $\lambda = 0.05$ to $+12.2$ at $\lambda = 10$ — a rigid model is *rewarded* — while the data-fit term falls from -0.75 near $\lambda = 0.45$ to -58.9 at $\lambda = 10$. Their sum peaks at $\lambda \approx 0.85$. That is an Occam's razor you did not have to write down, and it is a Lecture 1 optimisation nested inside the Lecture 4 loop.

Backup 3 — Expected Improvement, in closed form

With $f(x) \sim \mathcal{N}(\mu, \sigma^2)$ and incumbent f^+ , define $I = \max(0, f(x) - f^+)$. Integrate:

$$\text{EI}(x) = \int_{f^+}^{\infty} (f - f^+) p(f \mid \mathcal{D}) df = \sigma(x) \left[\underbrace{\frac{\mu - f^+}{\sigma} \Phi(z)}_{\text{exploit}} + \underbrace{\phi(z)}_{\text{explore}} \right], \quad z = \frac{\mu - f^+}{\sigma}$$

with Φ, ϕ the standard normal CDF and PDF. Adding a margin ξ gives the general form $\text{EI} = (\mu - f^+ - \xi)\Phi(Z) + \sigma\phi(Z)$, $Z = (\mu - f^+ - \xi)/\sigma$, and $\text{EI} = 0$ wherever $\sigma = 0$.

Reading the two terms. $(\mu - f^+)\Phi(z)$ is large where the mean already beats the incumbent; $\sigma\phi(z)$ is large where the uncertainty is high, even if the mean is unremarkable.

Why EI beats PI. PI integrates the *density* above the line — it counts whether an improvement happens. EI integrates the density *weighted by how far above the line it lands* — it counts how big. That single difference removes the need for a tuning parameter, and on the Act 3 posterior it moves the query from $x = 0.630$ (worth 1.075) to $x = 0.470$ (worth 1.301, against a true maximum of 1.303).

Backup 4 — the bandit toolkit, three lectures early

The source lecture develops the finite-armed bandit in full before reaching BO. Every rule below reappears in Part IV.

RULE	FORM	REAPPEARS AS
Action value	$Q_t(a) = \frac{r_1 + \dots + r_{k_a}}{k_a}$, greedy $a_t = \arg \max_a Q_t(a)$	the Q -function, Lecture 8
Incremental update	$Q_{k+1} = Q_k + \alpha_k [r_{k+1} - Q_k]$, with $\alpha_k = \frac{1}{k+1}$ or a constant	$new \leftarrow old + step \times (target - old)$ — every RL update, and a constant α gives exponential recency weighting
ε-greedy	$\pi(a) = 1 - \varepsilon + \frac{\varepsilon}{ A }$ if $a = a^*$, else $\frac{\varepsilon}{ A }$	Lecture 8's exploration rule, unchanged
Softmax	$\pi_t(a) = e^{Q_t(a)/\tau} / \sum_b e^{Q_t(b)/\tau}$	the Boltzmann policy; $\tau \rightarrow 0$ recovers greedy
Preference rules	reinforcement comparison $p_{t+1}(a_t) = p_t(a_t) + \beta[r_t - \bar{r}_t]$; pursuit, keeping both Q_t and π_t	REINFORCE with a baseline , and actor-critic — Lecture 10
UCB	$a_t = \arg \max_i (\mu_i + \sqrt{2 \ln t / n_i})$	$\mu(x) + \kappa \sigma(x)$, this lecture, continuous

Every arm is sampled infinitely often, so $Q_t \rightarrow Q^*$ and the chance of choosing the optimal action converges to at least $1 - \varepsilon$ — and never better, while ε is fixed. That floor is why the Act 4 curve turns back up.